# Hints: Linear algebra – 2. Vector spaces, dimension and rank

Class work

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

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## Problem 1

**Hint 1.** For the sum and the product, compare row spaces and column spaces: the rows of $A+B$ lie in row $A+$ row $B$, the columns of $AB$ lie in $\operatorname{im} A$, and the rows of $AB$ lie in the row space of $B$. For the block row, $\operatorname{im}(A \mid B) = \operatorname{im} A + \operatorname{im} B$.

**Hint 2.** Key lemma: restricting $A$ to $\operatorname{im} X$ and applying rank–nullity gives $\operatorname{rank}(AX) = \operatorname{rank} X - \dim(\ker A \cap \operatorname{im} X)$.

**Hint 3.** Apply the lemma with $X = B$ and with $X = BC$, and use $\ker A \cap \operatorname{im}(BC) \subseteq \ker A \cap \operatorname{im} B$: this is Frobenius. Sylvester is Frobenius with middle matrix $E_n$. Alternatively, the lemma with $X = B$ gives $\operatorname{rank} AB - (\operatorname{rank} A + \operatorname{rank} B - n) = \dim \ker A - \dim(\ker A \cap \operatorname{im} B)$, which is 0 exactly when $\ker A \subseteq \operatorname{im} B$.

**Hint 4.** For sharpness, use diagonal 0/1 matrices (coordinate projections onto index sets $S, T, \ldots$). Their ranks are $|S|, |T|$, and products correspond to intersections. Choose the sets so that the overlaps are as small as allowed, e.g. $A = \operatorname{diag}(E_a, 0)$, $B = \operatorname{diag}(0, E_b)$ give $\operatorname{rank} AB = \max(0, a + b - n)$.

## Problem 2

**Hint 1.** Reduce to a linearly independent subfamily $f_1, \ldots, f_k$; the intersection of kernels does not change.

**Hint 2.** Independence gives vectors $e_1, \ldots, e_k$ with $f_p(e_i) = \delta_{pi}$ (the map $V \to \mathbb{R}^k$, $x \mapsto (f_1(x), \ldots, f_k(x))$, is onto). For any $x$, the vector $y = x - \sum_j f_j(x) e_j$ lies in every $\ker f_p$, hence in $\ker f$. Read off the coefficients $f(e_j)$.

## Problem 3

**Hint 1.** Split a polynomial into homogeneous parts $P = \sum_k P_k$. The average of $P_k$ over the circle of radius $r$ is $r^k$ times its average over the unit circle. So $P$ is balanced if and only if every homogeneous part has zero average on the unit circle.

**Hint 2.** For odd $k$ the average is automatically 0 (since $P_k(-x, -y) = -P_k(x, y)$). For even $k$, "average $= 0$" is one non-trivial linear condition, because $(x^2 + y^2)^{k/2}$ has average 1.

**Hint 3.** So $\dim V = \binom{2011}{2} - 1005$, subtracting one condition for each even $k \in \{0, \ldots, 2008\}$. The answer is 2020050.

## Problem 4

**Hint 1.** $A = u\mathbf{1}^T + \mathbf{1}u^T$ with $u = (1, \ldots, n)^T$, so the rank is at most 2. Alternatively, subtract consecutive rows.

**Hint 2.** For $n \geqslant 2$ the two rows $(2, 3, \ldots)$ and $(3, 4, \ldots)$ are not proportional. The answer: rank 2 for $n \geqslant 2$, and 1 for $n = 1$.

## Problem 5

**Hint 1.**

(a) Write $A$ as a sum of $r$ rank-one matrices $u_k v_k^T$ (e.g. take a basis of the column space). Put the $u_k$ as the columns of $B$ and the $v_k^T$ as the rows of $C$. Then $\operatorname{rank} BC \leqslant \min(\operatorname{rank} B, \operatorname{rank} C) \leqslant r$ forces equality.

**Hint 2.**

(b) Let $D = CB \in M_r$. Then $A^{k+1} = BD^kC$. Take $p$ the characteristic polynomial of $D$ (degree $r$) and consider $q(x) = x\,p(x)$; Cayley–Hamilton gives $q(A) = Bp(D)C = 0$.

## Problem 6

**Hint 1.** Put $S = AB+BA$ and use $B^2 = B$. Compute $(B-E)S$, $S(B-E)$ and $(B-E)S(B-E)$.

**Hint 2.** You get $(B-E)S = BAB - AB$, $S(B-E) = BAB - BA$, and $(B-E)S(B-E) = 0$. Apply Frobenius' inequality to the triple $B-E$, $S$, $B-E$, and note that $AB-BA$ is the difference of the first two matrices.

## Problem 7

**Hint 1.** Show first that $f(PXQ) = f(X)$ for invertible $P, Q$: apply the inequality to $PXQ$ and to $X = P^{-1}(PXQ)Q^{-1}$.

**Hint 2.** Hence $f(X) = g(\operatorname{rank} X)$ with $g(r) = f(\operatorname{diag}(E_r, 0))$. From $J_r J_s = J_r$ for $r \leqslant s$, $g$ is non-decreasing, and it is surjective onto $\{0, \dots, n\}$. Hence $g = \mathrm{id}$: the only such function is $f(X) = \operatorname{rank} X$.

## Problem 8

**Hint 1.** Consider $S = A_1 + \cdots + A_k$. What does anticommutation do to $S^2$?

**Hint 2.** $S^2 = \sum A_i^2 + \sum_{i<j}(A_iA_j + A_jA_i) = S$, so $S$ is idempotent. For idempotents, rank equals trace, so $n \geqslant \operatorname{rank} S = \operatorname{tr} S = \sum \operatorname{rank} A_i$. Finish with pigeonhole.